

2(a)	any one from: suitable method for measuring a curve ; straighten shoot (before measuring) ;	1
2(b)(i)	38.7 <u>and</u> 45.3 ;	1
2(b)(ii)	difference is significant ;	1
2(c)	any two from: 1 ABA reduces (shoot) growth ; 2 SA increases (shoot) growth (by day 5) ; 3 ABA prevents, cell elongation / cell division or SA stimulates, cell elongation / cell division ; 4 description of the effect on growth by treatment C (ABA + SA) ; 5 ABA (partly) counteracts the effect (on growth) of SA or SA (partly) counteracts the effect (on growth) of ABA ;	3
2(d)	1 appropriate method of applying the solutions / hormones, to the roots only ; max. two from: 2 same volume of solutions ; 3 measuring shoot lengths on day 1 and day 5 ; 4 <u>named</u> example of condition which is <u>same</u> as shoot spraying experiment ; 5 method to identify presence of hormone in shoot ;	3

Question	Answer	Marks														
2(e)(i)	<i>one mark per row</i> <table border="1"> <thead> <tr> <th rowspan="2">gene</th><th colspan="2">effect on gene expression</th></tr> <tr> <th>ABA alone</th><th>SA alone</th></tr> </thead> <tbody> <tr> <td><i>CDKB2</i></td><td>large decrease</td><td>no change / small increase ;</td></tr> <tr> <td><i>KRP4</i></td><td>large increase</td><td>no change / small increase ;</td></tr> <tr> <td><i>CycD6</i></td><td>large decrease</td><td>no change / small decrease ;</td></tr> </tbody> </table>	gene	effect on gene expression		ABA alone	SA alone	<i>CDKB2</i>	large decrease	no change / small increase ;	<i>KRP4</i>	large increase	no change / small increase ;	<i>CycD6</i>	large decrease	no change / small decrease ;	3
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2(e)(ii)	<i>conclusion 1</i> fully supports <u>and</u> mRNA (ABA + SA / batch C) lower than with ABA (batch B) / ora ; <i>conclusion 2</i> partially supports <u>and</u> mRNA (ABA + SA / batch C) higher than with ABA (batch B) but not as high as with water (batch A) / SA (batch D) or no statistical <u>test</u> carried out ;	2														